

```

%% ===== 1. INITIALIZATION =====

clc; clear; close all;

files = dir('*.mat');

numFiles = length(files);

features = [];

labels = [];

%% ===== 2. FEATURE EXTRACTION =====

for i = 1:numFiles

    data = load(files(i).name);

    img = data.cjdata.image;

    label = data.cjdata.label;

    % ----- Preprocessing -----

    img = mat2gray(img);

    img = imresize(img,[256 256]);

    img = adapthisteq(img);

    img = medfilt2(img,[3 3]);

    % ----- Segmentation (using mask) -----

    mask = imresize(data.cjdata.tumorMask,[256 256]);

    mask = mask > 0.5;

    tumor = img .* mask;

    % ----- Feature Extraction (GLCM) -----

    tumor_gray = im2uint8(mat2gray(tumor));

    glcm = graycomatrix(tumor_gray, ...
        'Offset',[0 1; -1 1; -1 0; -1 -1], ...
        'Symmetric',true);

```

```

stats = graycoprops(gldcm, {'Contrast','Correlation','Energy','Homogeneity'});

contrast = mean(stats.Contrast);
correlation = mean(stats.Correlation);
energy = mean(stats.Energy);
homogeneity = mean(stats.Homogeneity);

entropyVal = entropy(tumor_gray);

features = [features;
            contrast correlation energy homogeneity entropyVal];

labels = [labels; label];
end

%% ===== 3. NORMALIZATION =====
features = normalize(features);

%% ===== 4. TRAIN-TEST SPLIT =====
cv = cvpartition(labels,'HoldOut',0.3);
X_train = features(training(cv),:);
Y_train = labels(training(cv));
X_test = features(test(cv),:);
Y_test = labels(test(cv));

%% ===== 5. TRAIN MODEL =====
model = fitcecoc(X_train, Y_train); % Multi-class SVM

%% ===== 6. TEST MODEL =====
predicted = predict(model, X_test);

```

```
accuracy = sum(predicted == Y_test)/length(Y_test)*100;
fprintf('Test Accuracy = %.2f%%\n', accuracy);
```

```
figure
confusionchart(Y_test, predicted);
title('Confusion Matrix');
```

```
%% ===== 7. SINGLE IMAGE PREDICTION =====
```

```
% Change file name for demo
testData = load('3000.mat');
```

```
img = testData.cjdata.image;
```

```
% Preprocessing
img = mat2gray(img);
img = imresize(img,[256 256]);
img = adapthisteq(img);
img = medfilt2(img,[3 3]);
```

```
% Segmentation
mask = imresize(testData.cjdata.tumorMask,[256 256]);
mask = mask > 0.5;
tumor = img .* mask;
```

```
% Feature extraction
tumor_gray = im2uint8(mat2gray(tumor));
```

```
glcm = graycomatrix(tumor_gray, ...
    'Offset',[0 1; -1 1; -1 0; -1 -1], ...
```

```
'Symmetric',true);
```

```
stats = graycoprops(gldm, {'Contrast','Correlation','Energy','Homogeneity'});
```

```
testFeature = [mean(stats.Contrast), ...  
               mean(stats.Correlation), ...  
               mean(stats.Energy), ...  
               mean(stats.Homogeneity), ...  
               entropy(tumor_gray)];
```

```
testFeature = normalize(testFeature);
```

```
% Prediction
```

```
predicted_label = predict(model, testFeature);
```

```
fprintf('Predicted Tumor Class = %d\n', predicted_label);
```

```
% Display
```

```
figure
```

```
subplot(1,2,1), imshow(img), title('Preprocessed MRI')
```

```
subplot(1,2,2), imshow(tumor), ...
```

```
title(['Segmented Tumor - Class: ', num2str(predicted_label)])
```